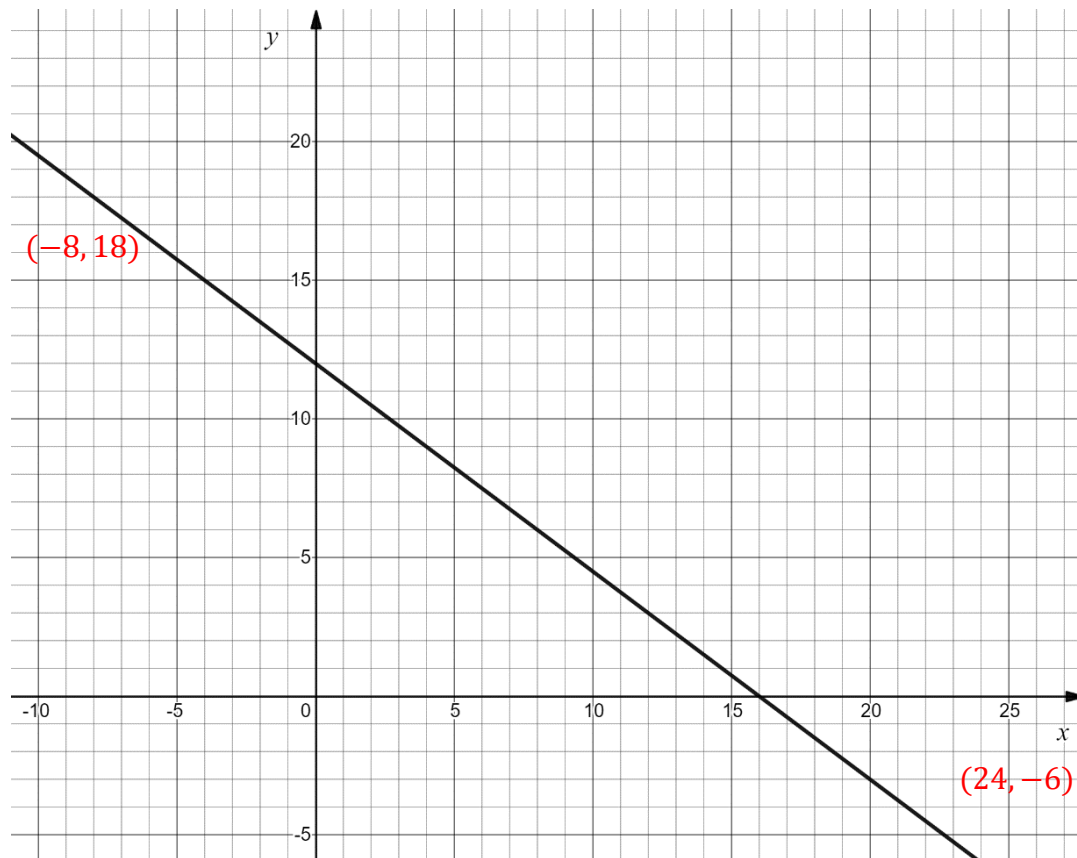


Algebra 1
Unit 4
Lesson 7 Practice

Name **Answer Key** _____
Date _____
Period _____

The function $f(x) = -\frac{3}{4}x + 12$ is shown on the coordinate grid.



1. Use the graph to find the solution $f(x) = 18$. Label the point that represents the solution.

$(-8, 18)$

2. Solve $-\frac{3}{4}x + 12 = 18$. Show your work.

$$-\frac{3}{4}x + 12 = 18$$

$$-12 \quad -12$$

$$-\frac{3}{4}x = 6$$

$$x = -\frac{24}{3}$$

$$x = -8$$

3. Use the graph to find the solution $f(x) = -6$. Label the point that represents the solution.

$(24, -6)$

4. Solve $-\frac{3}{4}x + 12 = -6$. Show your work.

$$-\frac{3}{4}x + 12 = -6$$

$$\quad -12 \quad -12$$

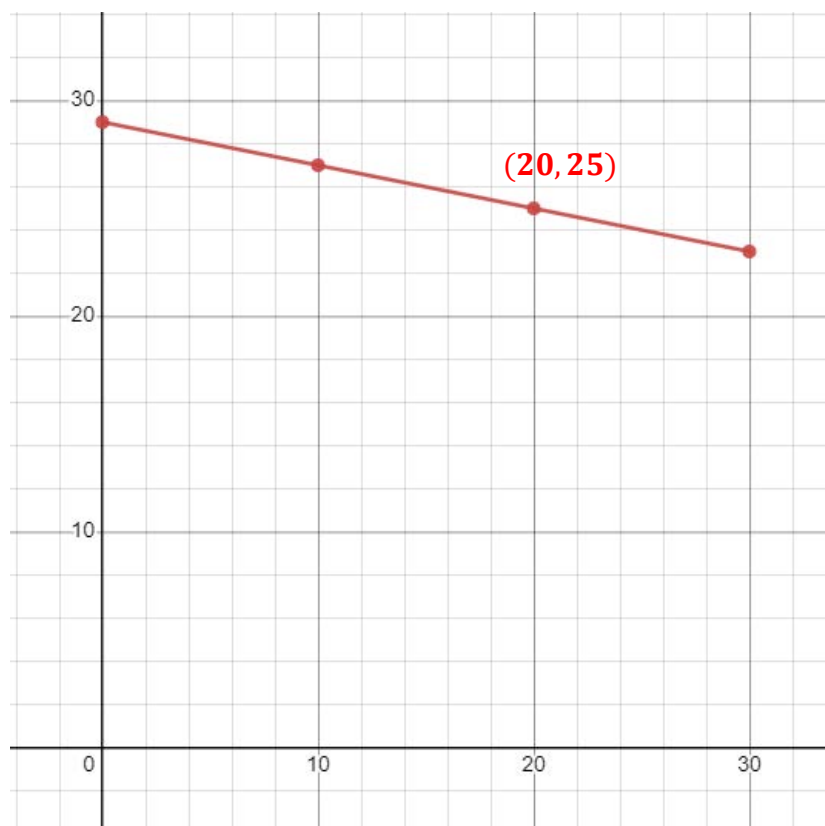
$$-\frac{3}{4}x = -18 \cdot -\frac{4}{3}$$

$$x = \frac{72}{3}$$

$$x = 24$$

From 1990 to 2020, in Nigeria the percentage of land that are forests has declined. The function $F(t) = -0.2t + 29$ models the percentage of the land, F , that is forested where t is the number of years since 1990.

5. Explain why the values of t should have a constraint.
 t should have a constraint because the equation only applies to the years between 1990 and 2020. There can't be negative years since 1990.
6. Write the constraint for the values of t using set builder notation.
 $\{t | 0 \leq t \leq 30\}$
7. Write the constraint for the values of F using set builder notation.
 $\{F | 23 \leq F \leq 29\}$
8. Graph the function using the constraints.



9. Use the graph to find the solution $F(t) = 25$. Label the point that represents the solution.

$$25 = -0.2t + 29$$

$$-29 \quad -29$$

$$\frac{-4}{-0.2} = -\frac{0.2t}{-0.2}$$

$$t = 20$$

10. Solve $0.5t + 47 = 55$. Show your work.

$$-47 \quad -47$$

$$\frac{0.5t}{0.5} = \frac{8}{0.5}$$

$$t = 16$$